



Guidelines for performing mutational signature analysis

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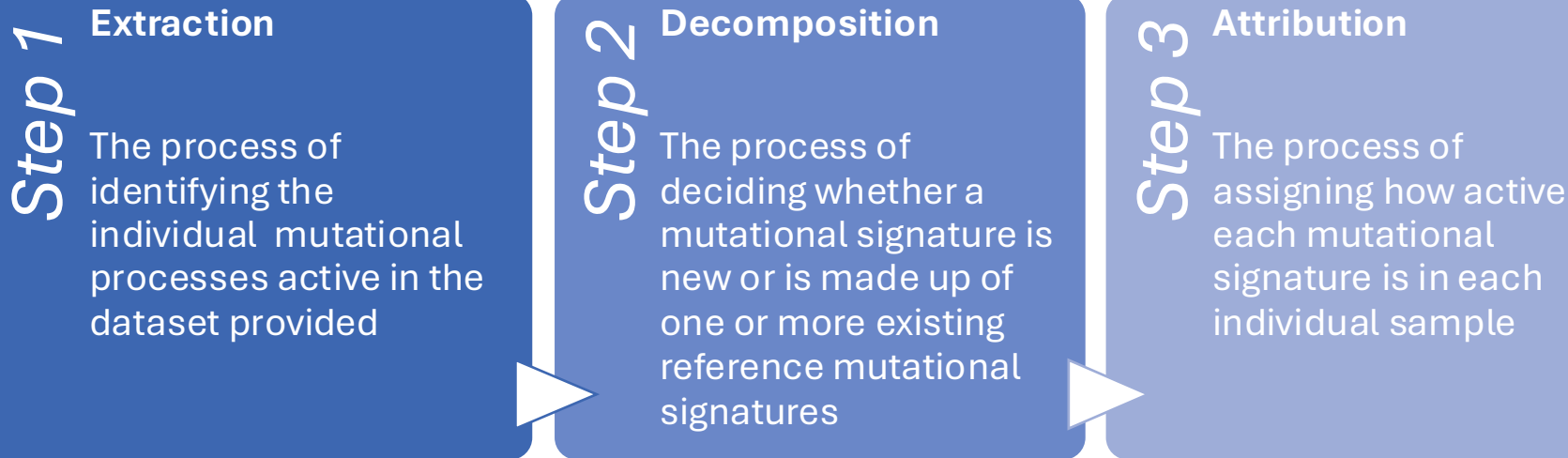
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Mutational Signature Analysis

Mutational Signature Analysis can be split into 3 distinct phases:



There are now **many** available methods for mutational signature analysis
Regardless of which tools used, the results need to be manually reviewed
There are some common principles which may be helpful

Step 1: Mutational Signature Extraction

SigProfilerExtractor

SBS96 / SBS288 / SBS1536
DBS78 and ID83

HDP

SBS96 / SBS288
DBS78 and ID83

Can be run with priors but
this is not recommended

Hierarchies can be used,
with multiple tiers

OR

mSigHdp

SBS96 / SBS288
DBS78 and ID83

Does not run with priors

Single tier Hierarchy only

Note: Hierarchy

It is recommended to test with and without the hierarchy

Note: Subsetting

If your dataset contains
samples with MSI or other
DNA repair deficiencies, you
should:

a) Perform extractions only
on the subset of cases
without such features

OR

b) Run a second extraction
to confirm that there are no
signatures which can only
be extracted when such
samples are absent

*If extractions are being
performed on multiple
tissues it is also
recommended to perform
tissue specific extractions
as a check*

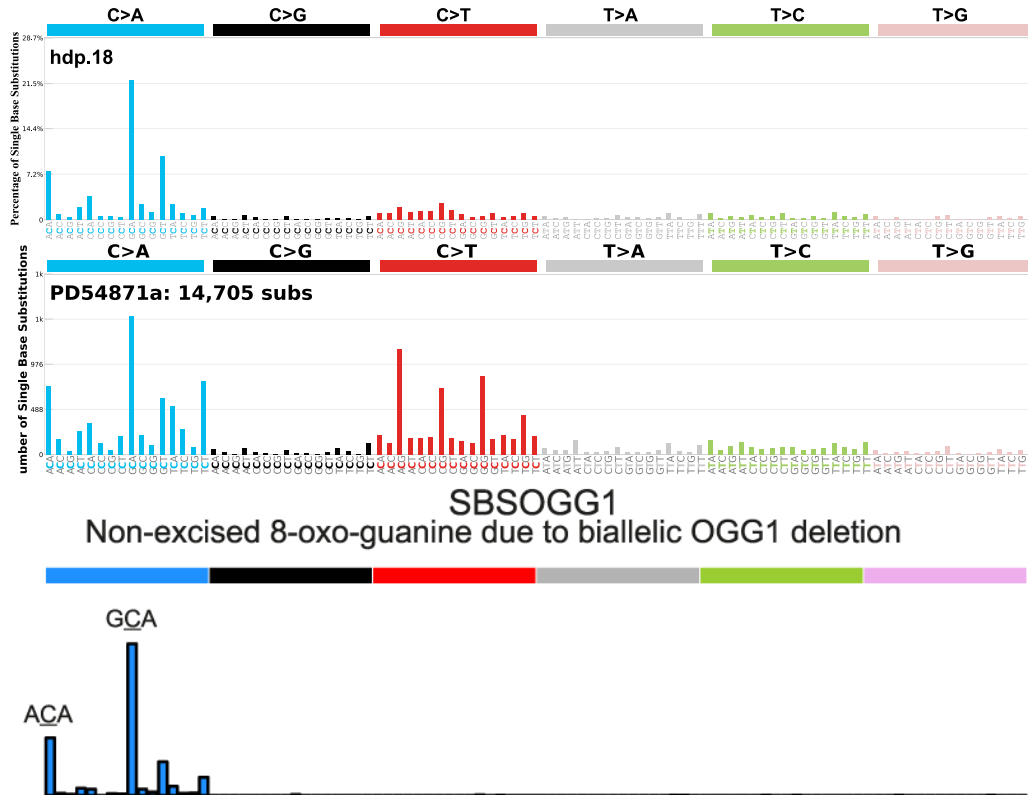
Compile consensus *de novo* signature list

Reasons to accept signatures:

- Are present in multiple signature extractions (as independent or combined signatures)
 - Are clearly supported by individual mutational spectra
 - Represent known COSMIC signatures

Step 2: Mutational Signature Decomposition

Examples – Multiple algorithms

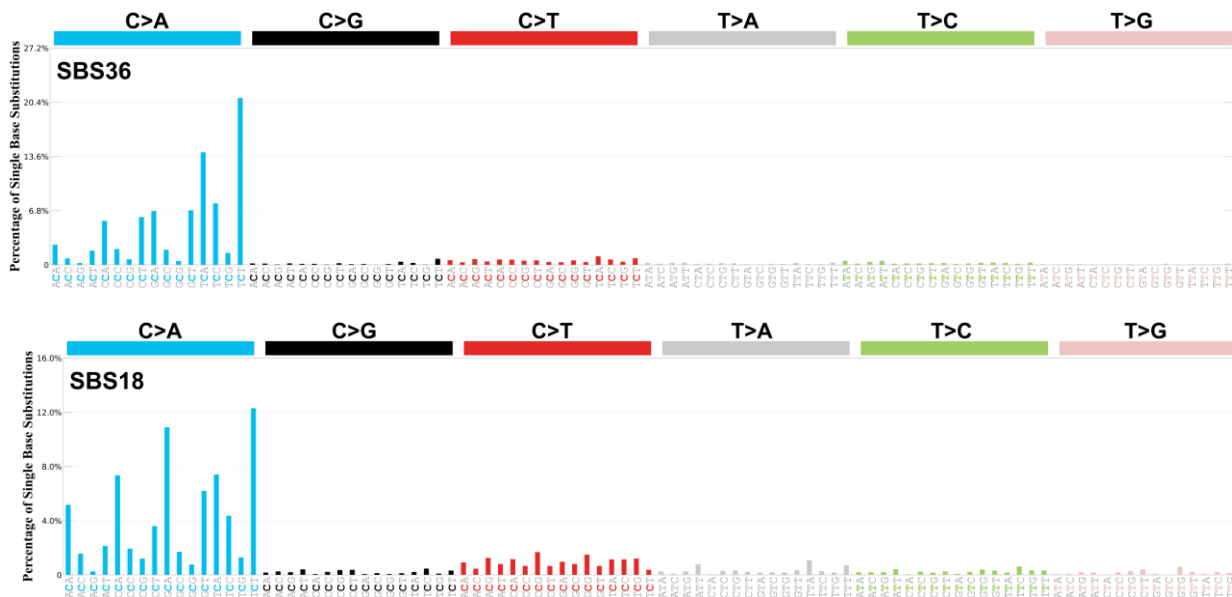


Running multiple algorithms is useful:

SigprofilerExtractor missed this signature which originates from a single case with a germline OGG1 variant

Examples - Subsetting

An example of a signature we expect to be present being absent from the results

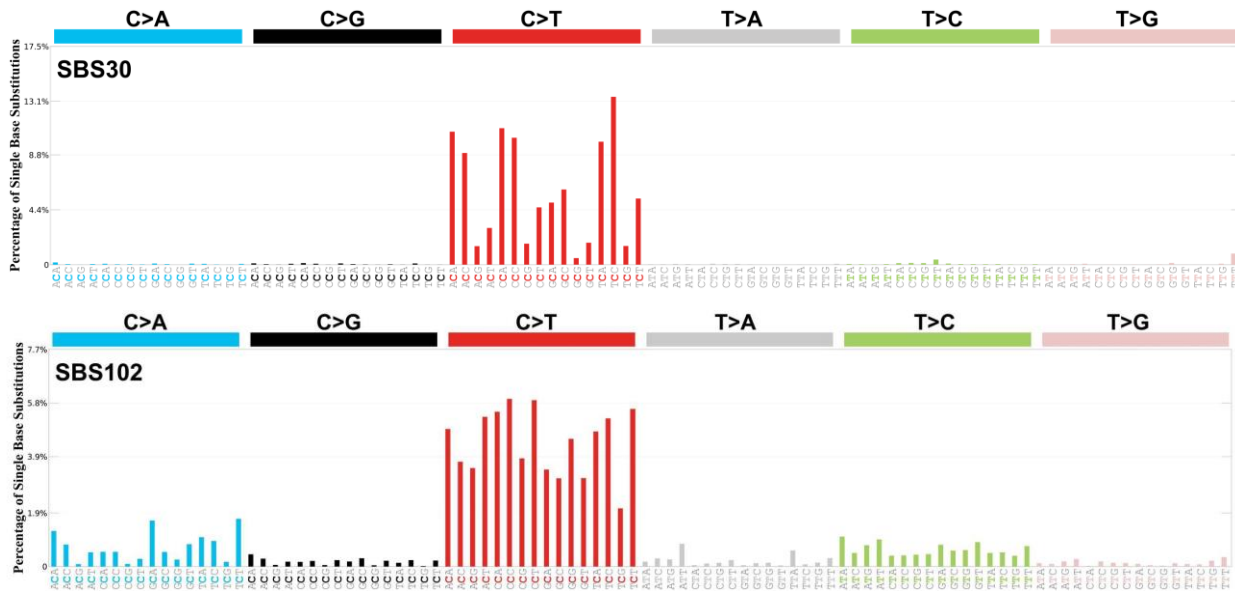


SBS36 is a signature caused by inactivation of MUTYH, resulting in defective base excision repair

The presence of samples with SBS36 very often blocks SBS18, which is due to damage from reactive oxygen species found in many tissues

Examples - Subsetting

An example of a novel signature with interesting properties being blocked

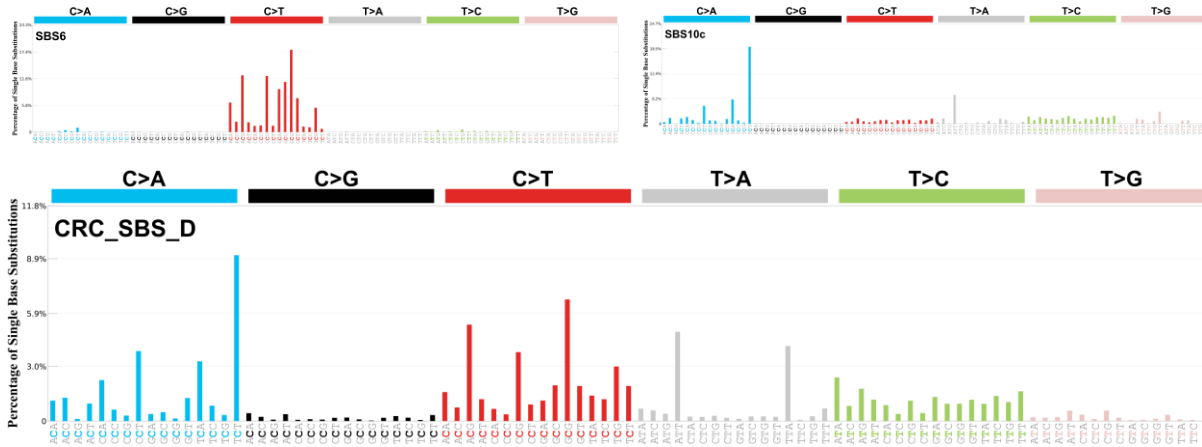


SBS30 is a signature caused by inactivation of NTHL1, resulting in defective base excision repair

The presence of samples with SBS30 would block the extraction of a novel signature found in colorectal cancers which displayed geographic variation

Examples - Subsetting

An example of a novel signature with interesting properties being blocked

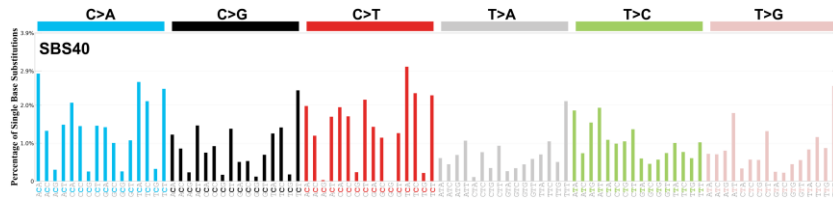


MSI and Polymerase deficiency signatures generate very large numbers of samples

The presence of these signatures prevents the discovery of a novel signature which has many interesting features

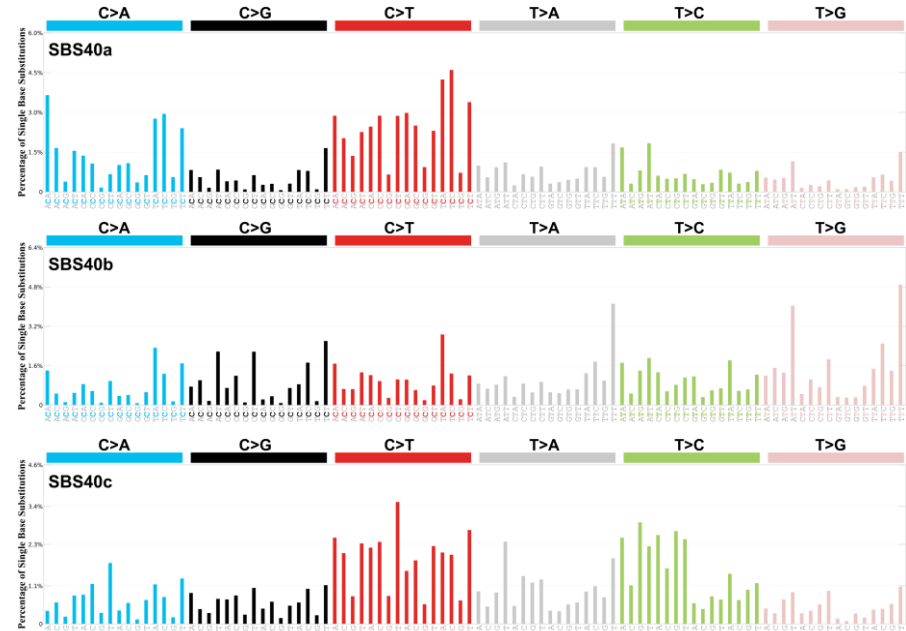
Example – Tissue Specific Extractions

PCAWG – Multiple Tissues



Evidence for this split can be seen in the tissue specific results in the PCAWG supplementary materials

Mutographs Project – Kidney only



Step 2: Mutational Signature Decomposition

Decompose Signatures

SigProfilerAssignment

Default Parameters

- COSMIC references only
- No subgroups excluded
- Threshold = 0.8 CS

Optimised Parameters

- Can include non-reference signatures
- Exclude appropriate subgroups

Cosine Similarity (CS) to COSMIC reference signatures

Compile final COSMIC plus *de novo* signature list:

Justifications to reject default decompositions and consider a signature novel:

- If the default decomposition includes biologically implausible signatures
- Individual mutational spectra which support the existence of the signature
- If particular peaks / mutation classes are not well explained by the decomposition
- Signature is associated with metadata/risk factors or in exposure studies, found only in the treated subset
- For flat signatures, requiring a higher threshold for CS can be justified as flat signatures can be more readily reconstructed with existing signatures

Note: Optimised Decompositions

When excluding signature subgroups, the following are examples of justifications:

- Review of mutational spectra
- Use of subsets for extractions
- Absence of mutations which generate specific signatures (e.g. NTHL1 and SBS30)
 - Tissue Type
 - Previous Literature

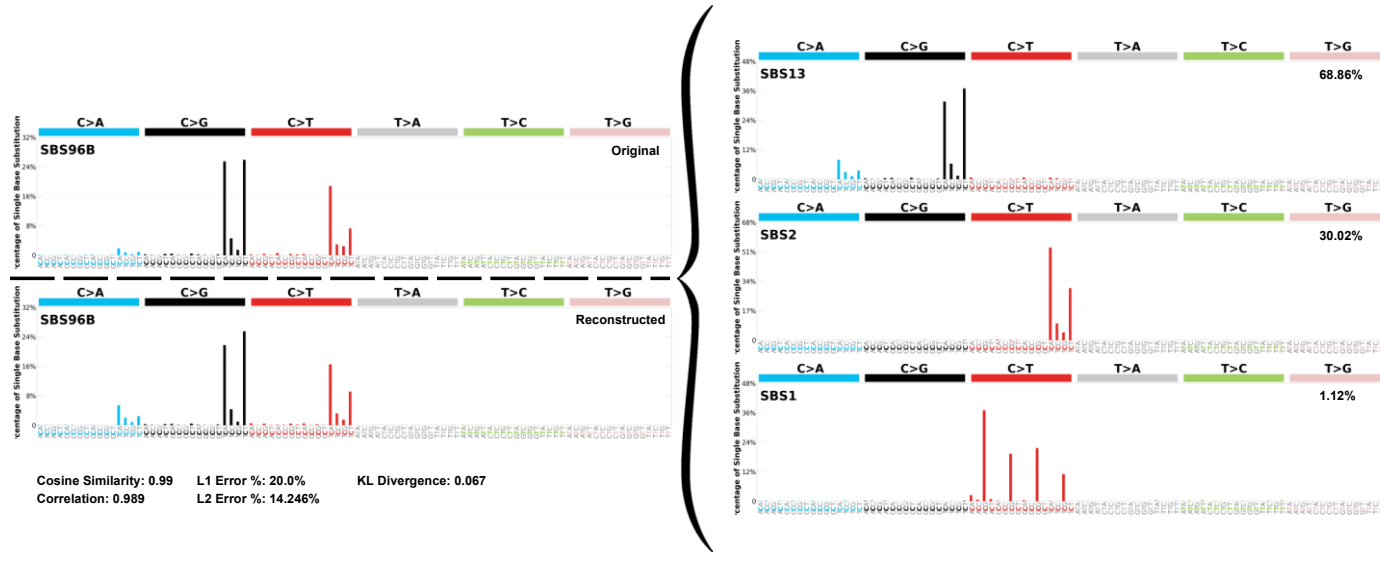
It is important to review both default and optimised decompositions

Step 3: Mutational Signature Assignment

Examples - Decomposition

Decomposition plots and stats need to be checked thoroughly
It is very common to see signatures used inappropriately

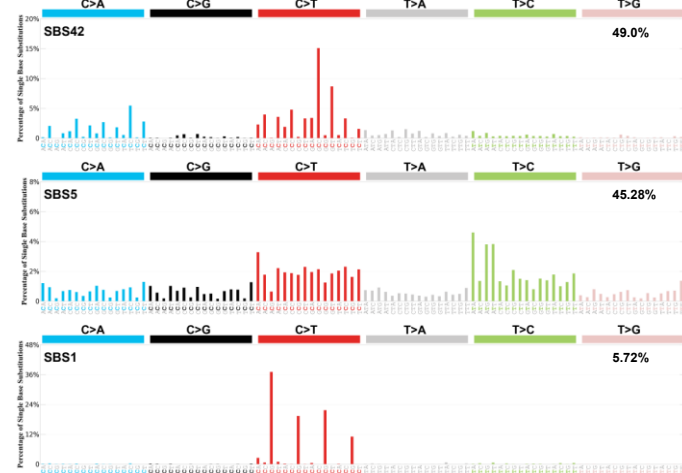
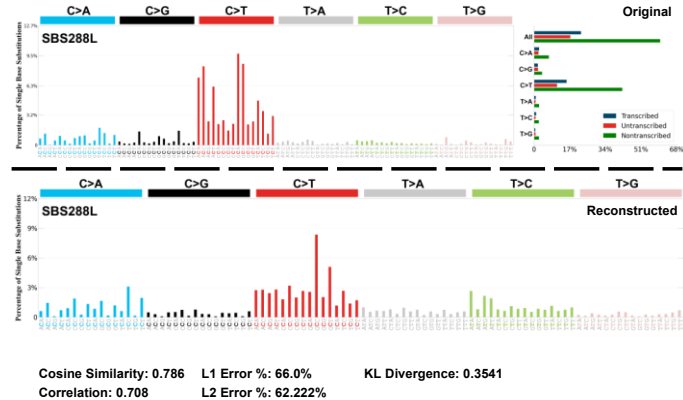
Example 1: A signature that should be decomposed



Examples - Decomposition

Decomposition plots and stats need to be checked thoroughly
It is very common to see signatures used inappropriately

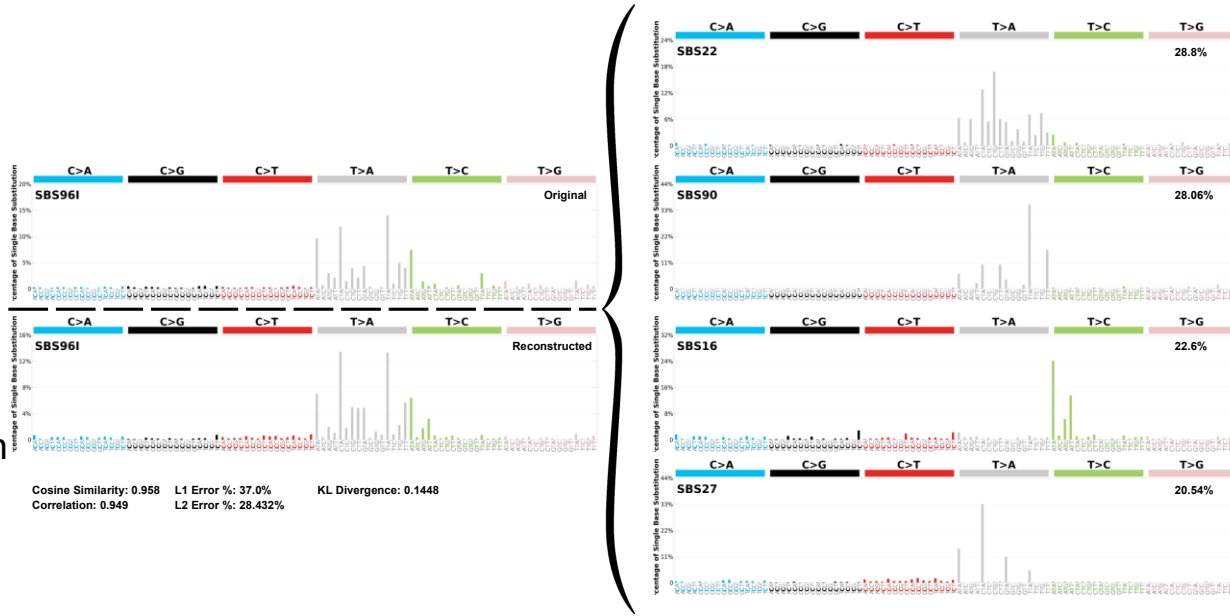
Example 2: A signature that should NOT be decomposed (and is not)



Examples - Decomposition

Decomposition plots and stats need to be checked thoroughly
It is very common to see signatures used inappropriately

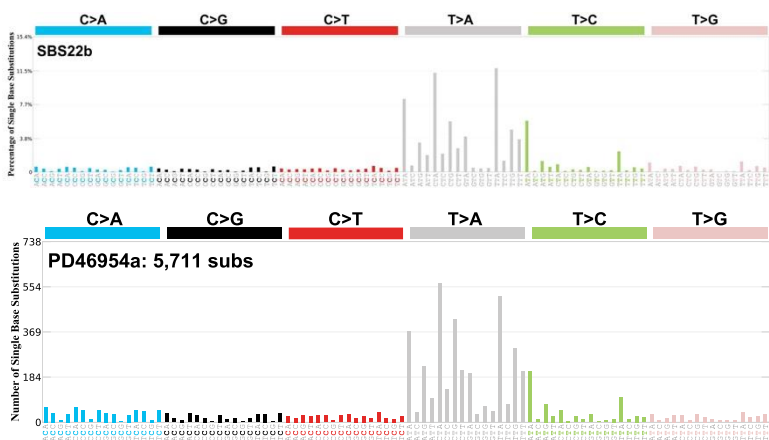
Example 3: A signature that should NOT be decomposed (but is)



SBS90 is a
chemotherapy
signature

SBS27 is an artifact
signature

Examples - Justification for Novel Signatures



This signature was deemed to be novel because:

- Presence of individual samples with strong support for the new signature
- Extracted using multiple independent extractions
- Decomposition using existing reference signatures is biologically implausible
- Linked to Aristolochic Acid exposure: correlates with experimental validated signatures (SBS22a) and similar pattern of geographic variation

Step 3: Mutational Signature Assignment

Signature Assignment

Tier 1:

SigProfilerAssignment
/Extractor results

Hdp / mSigHdp results

*If there are concerns about a
given signature, it is
recommended to run tests with
and without the signature to
assess impact*

Tier 2:

MSA

OR

mmSig

Signatures
extracted from
single tissue

Signatures extracted from multiple tissues
OR assigning all COSMIC signatures

Step1 :Tissue specific attribution with all signatures

Step2 :Tissue specific attribution with ONLY signatures
present after pruning in step 1 results

Final assignments for downstream analyses

Downstream Analyses

Note: Assignment Algorithms Classification:

Tier 1 algorithms

- Quick and Easy
- No/limited optimisation
 - No pruning

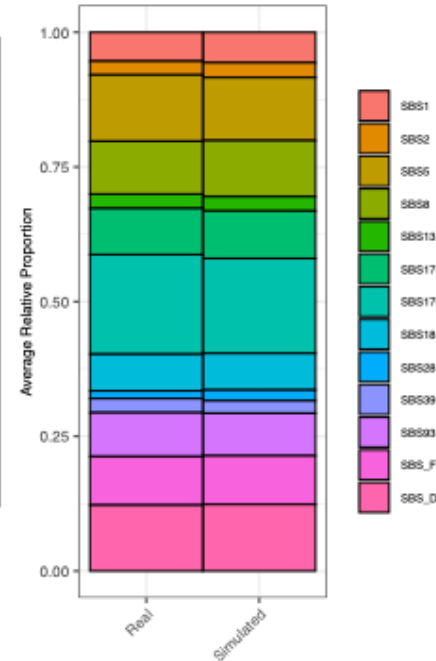
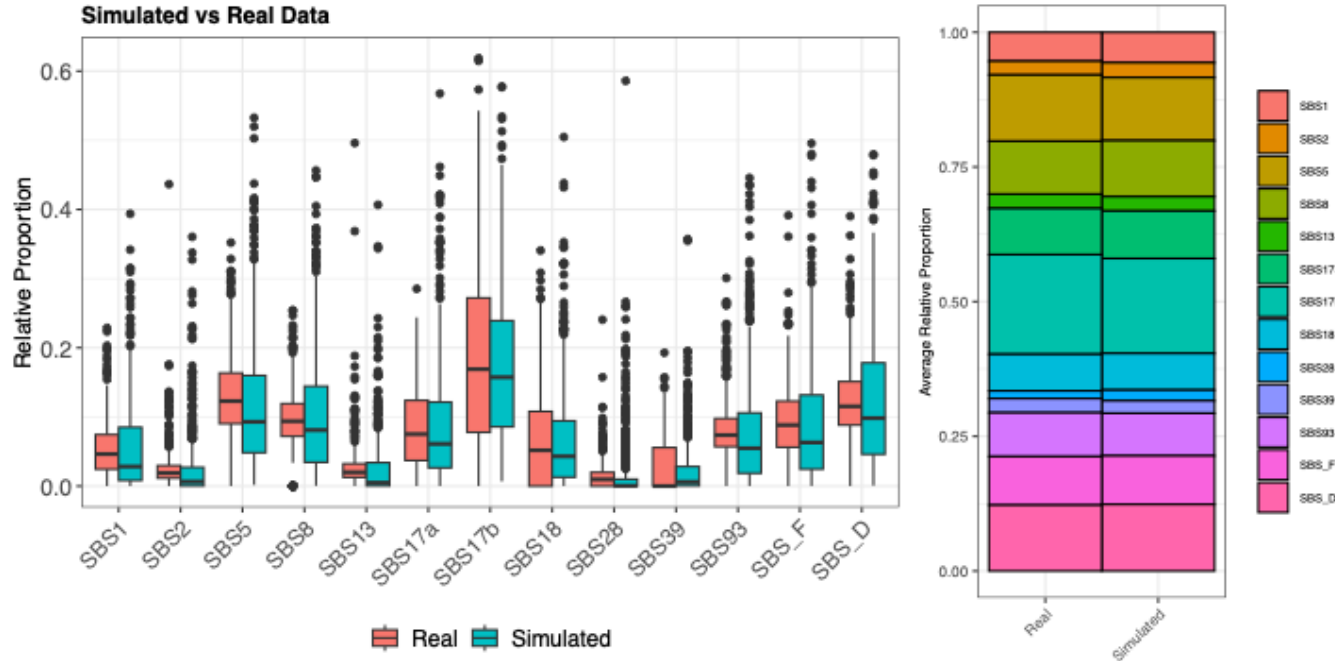
*Use For: initial results and
quick tests*

Tier 2 algorithms

- Resource intensive
- Optimisation
- Prunes attributions
 - Published &
benchmarked

Use For: final results

Examples: Simulations

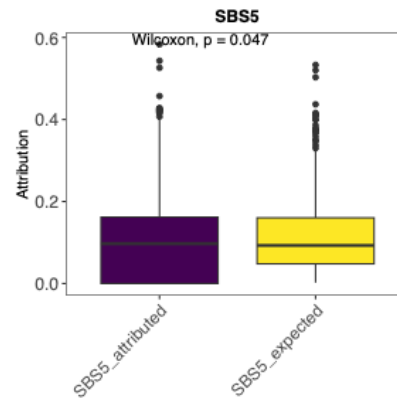
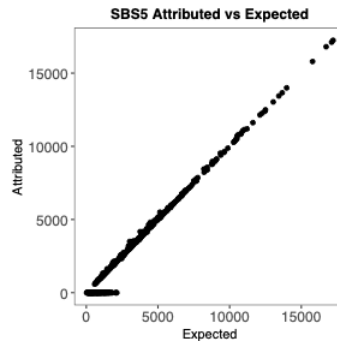
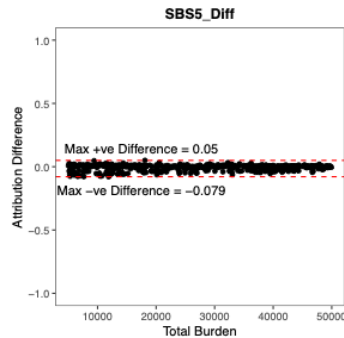


Recently, we have started performing tests using simulated datasets to validate the attribution of mutational signatures

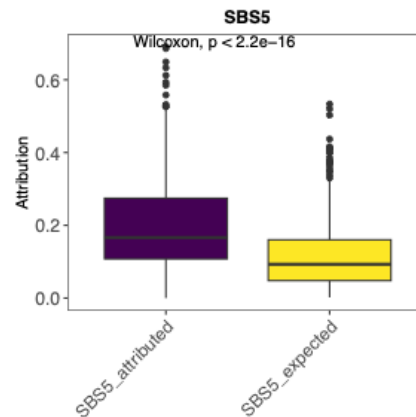
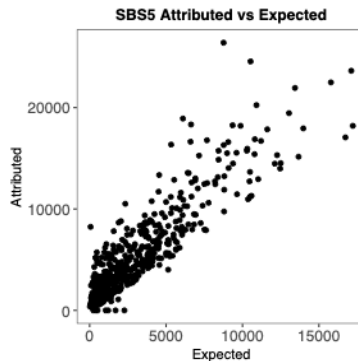
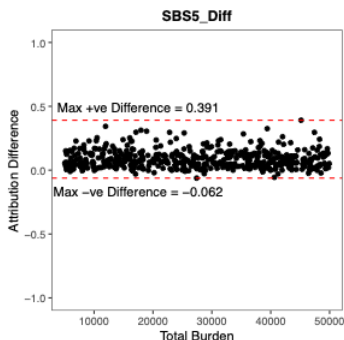
The simulated dataset is based on what we believe to be true – but still allows us to assess how well different attribution methods function

Examples - Simulations

MSA



SPA



MSA performs much better for flat signatures such as SBS5

SPA tends to significantly overestimate the level of SBS5